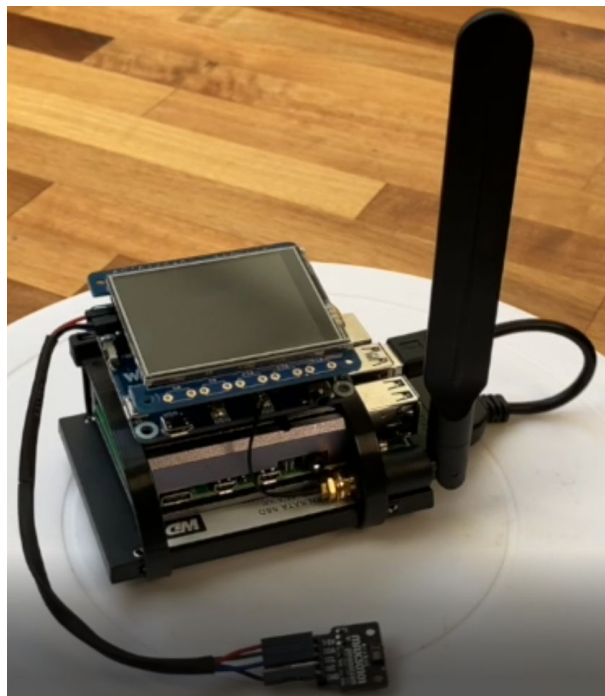


End to End IoT with Dart and Flutter

Flutter Vikings - August 2022



A brief introduction



Engineer, Atsign & Dart GDE
building a platform that puts people in
control of their data.

Co-host, Tech Debt Burndown Podcast

Cloud Editor, InfoQ

Blogger blog.thestateofme.com

chris.swanz.net for links to socials etc.



@cpswan



Agenda

- Overview of Mobile World Congress Demo
- The Flutter front end
- A Dart device driver
- Connectivity, Security, Privacy
- Local Display
- Review



Overview of Mobile World Congress Demo



All the code is open source

atsign-foundation / mwc_demo Public

generated from atsign-company/archetype

<> Code Issues Pull requests Actions Projects Wiki Security Insights Settings

trunk 8 branches 4 tags

Go to file Add file Code

 cpswan Punctuation bd9e731 13 days ago 131 commits

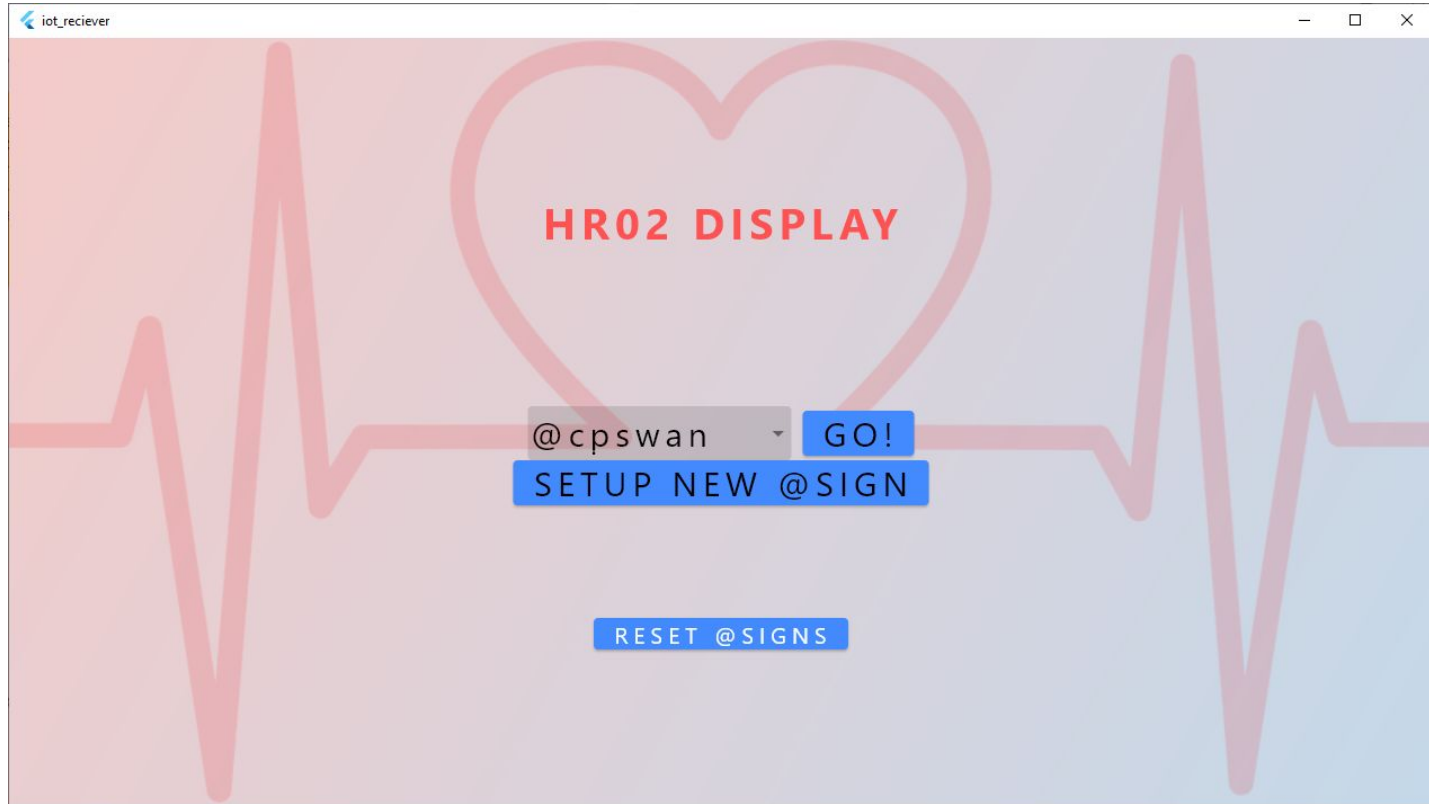
.github	Initial commit	last month
at_cli	Script to read HR from MQTT and publish to @sign	last month
dart/iot_sender	Split out max30101 package into its own repo	14 days ago
flutter/iot_reciever	apple version in store	29 days ago
gsm	add ppp.sh	15 days ago
python	More detailed message	15 days ago
shell	Reduce range	last month
systemd	systemd units	15 days ago
.gitignore	Remove hive files and update .gitignore	last month
CONTRIBUTING.md	correct repo name etc.	15 days ago
LICENSE	Initial commit	last month
README.md	Punctuation	13 days ago
code_of_conduct.md	Initial commit	last month



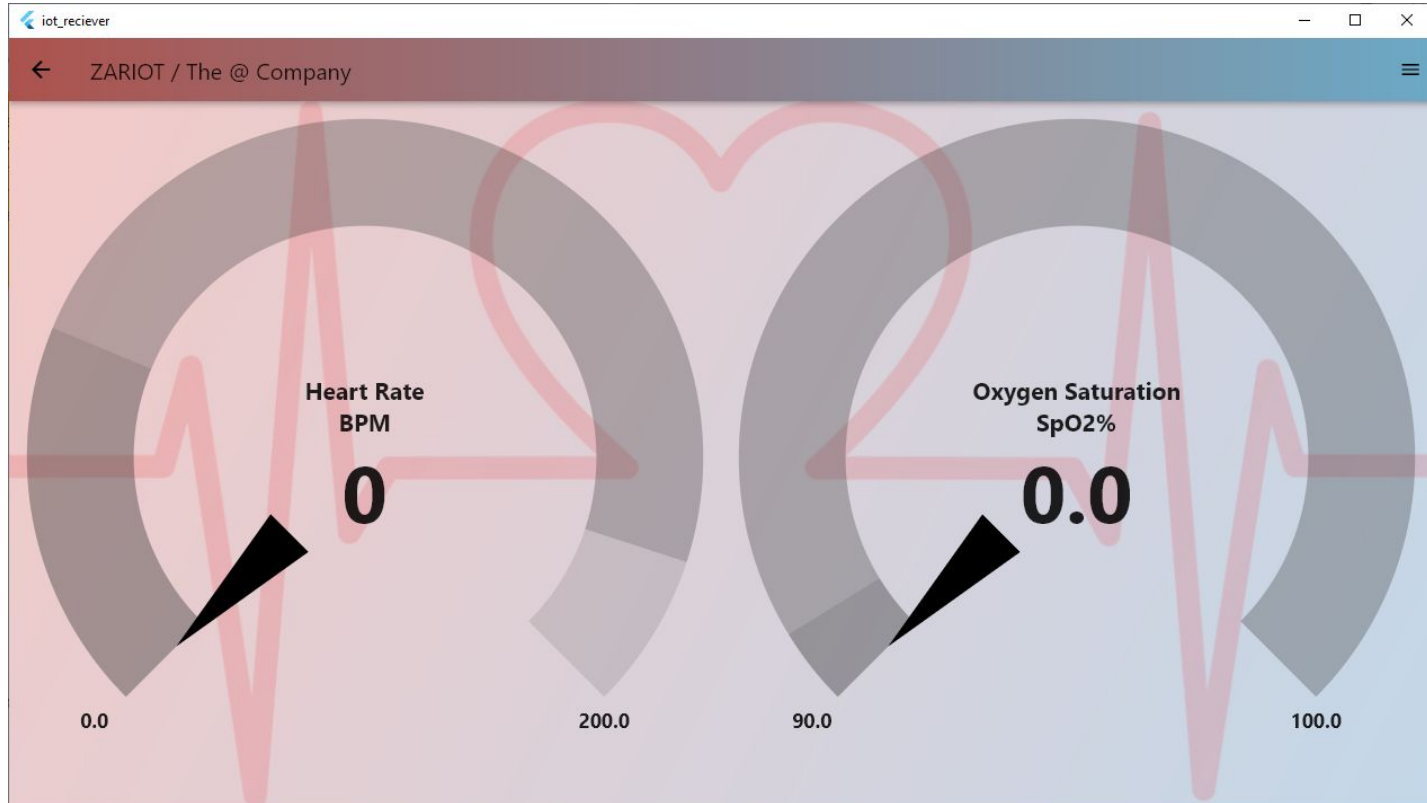
https://github.com/atsign-foundation/mwc_demo

Flutter Front End

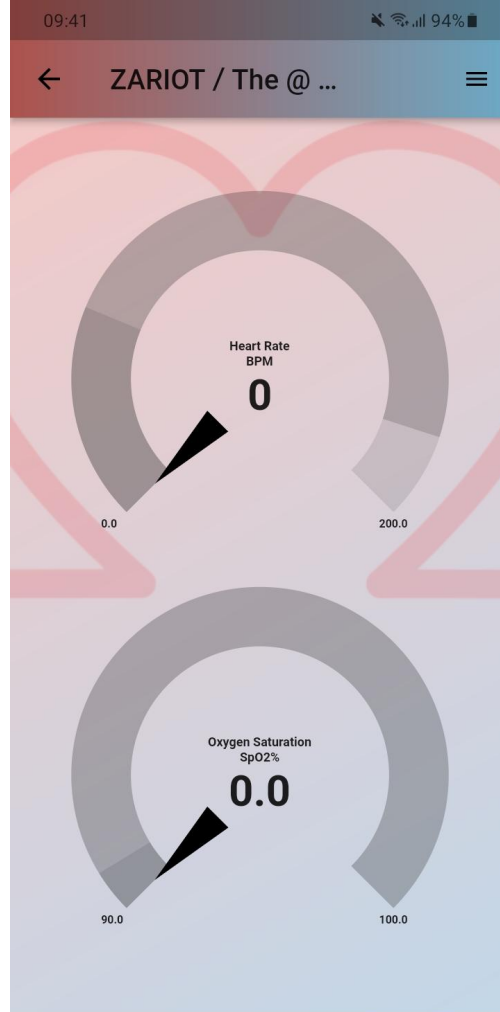
Sign in (Windows Desktop)



Main screen (Windows Desktop)



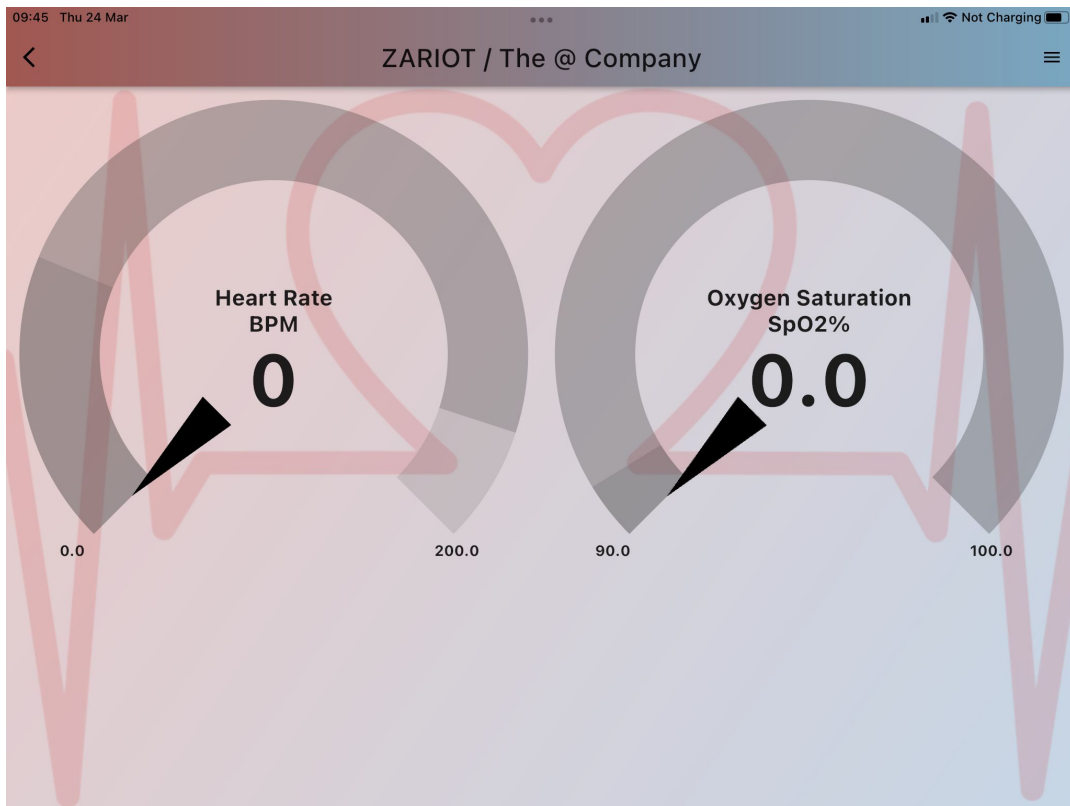
And Android



<https://play.google.com/store/apps/details?id=com.atsign.hro2>



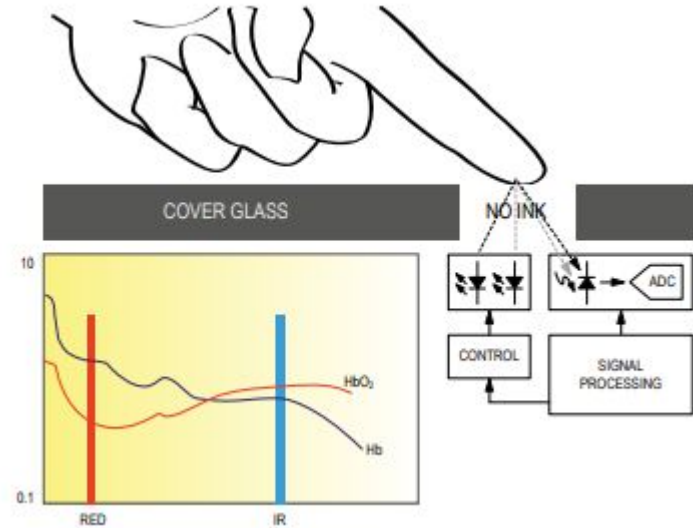
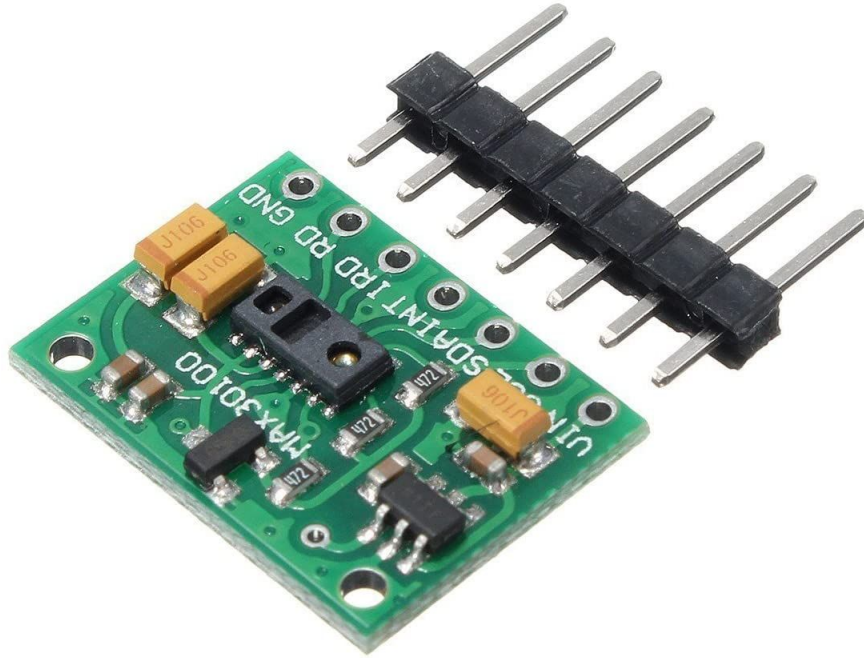
And iOS



<https://apps.apple.com/gb/app/hro2/id1610811005>

Dart Device Driver

MAX30100 sensor - heart rate and O2 saturation



The programmer's lament

*“We do these things,
not because they are easy,
but because we thought they would be easy”*



Newer sensor - better drivers?



<https://shop.pimoroni.com/products/max30101-breakout-heart-rate-oximeter-smoke-sensor>



C++ (and bits of Python) driver ported to Dart

max30101 1.0.0-RC3

Published 12 days ago • [atsign.org](#) Null safety

SDK

DART

FLUTTER

PLATFORM

ANDROID

IOS

LINUX

MACOS

WINDOWS

👍 0

[Readme](#)

[Changelog](#)

[Example](#)

[Installing](#)

[Versions](#)

[Scores](#)

[Admin](#) 🔒

[Activity log](#) 🔒

max30101

A Dart package (developed for use on a Raspberry Pi 4) which interfaces via I2C with the [Max30101](#) sensor and produces illustrative heart rate and O2 saturation readings.

- Primarily based on Raivis Strogonovs' excellent [C++ library](#) which implements heart rate and O2 saturation readings for the Max30100 sensor. (See also his superb accompanying [tutorial](#)). This package is a pretty faithful port to Dart of that C++ code, and also includes the various adjustments (registers, settings, etc.) required to make it work with the Max30101 sensor.
- The package's approach to register interaction was inspired by Philip Howard's [Python library](#) which implements heart rate readings for the Max30105 sensor
- The essential ingredient of I2C support is provided by the https://pub.dev/packages/dart_periphery package



<https://pub.dev/packages/max30101>



We stand on the shoulders of giants

dart_periphery 0.8.29

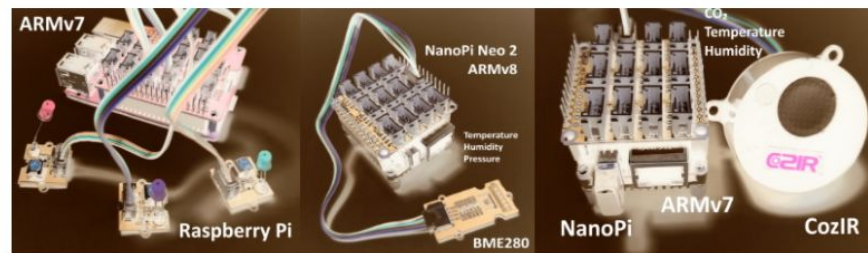
Published 4 days ago • flutterdev.at (Null safety)

SDK | DART | FLUTTER | PLATFORM | ANDROID | IOS | LINUX | MACOS | WINDOWS

18

[Readme](#) | [Changelog](#) | [Example](#) | [Installing](#) | [Versions](#) | [Scores](#)

dart_periphery



pub v0.8.29

Introduction

dart_periphery is a Dart port of the native [c-periphery library](#) for Linux Peripheral I/O (GPIO, LED, PWM, SPI, I2C, MMIO and Serial peripheral I/O). This package is specially intended for SoCs like Raspberry Pi, NanoPi, Banana Pi et al.



https://pub.dev/packages/dart_periphery

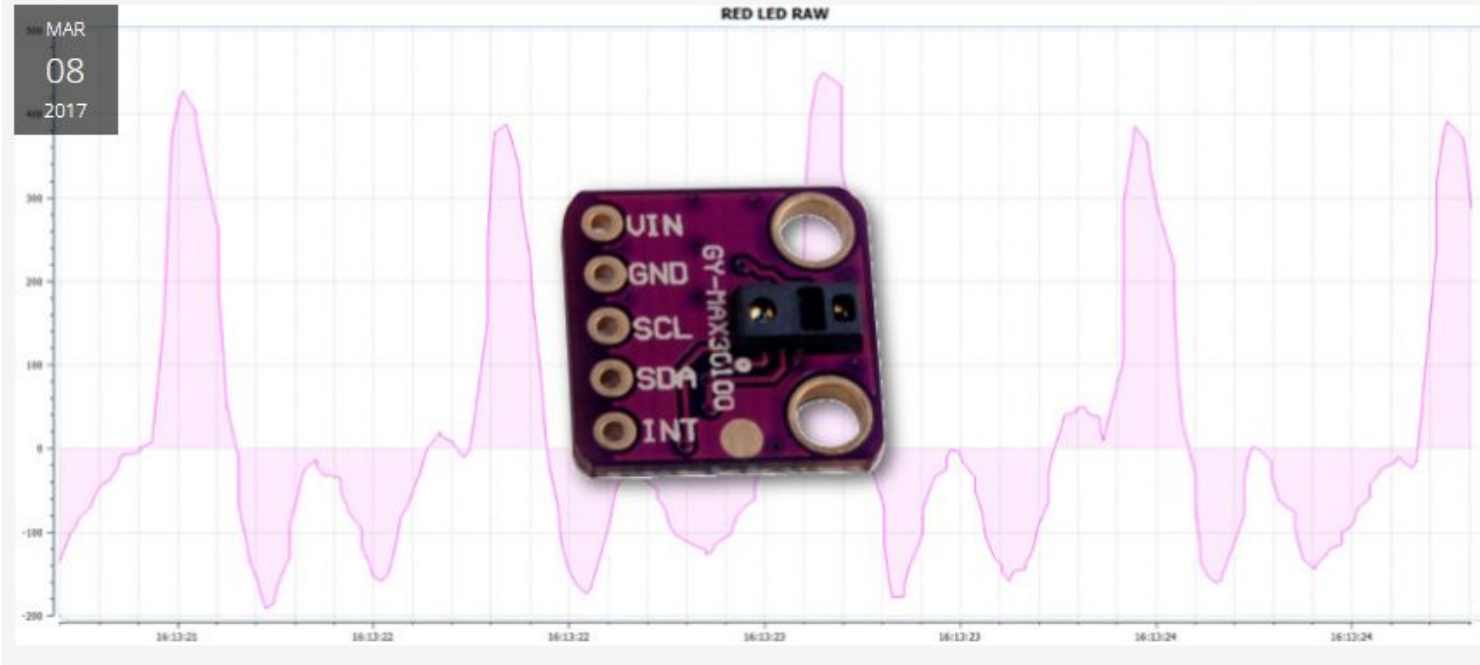


C++ code is EXCEPTIONALLY well documented

Implementing pulse oximeter using MAX30100

by Raivis Stroganovs In: Microcontrollers, Atmel

max30100, pulse oximeter, arduino 120 Comments

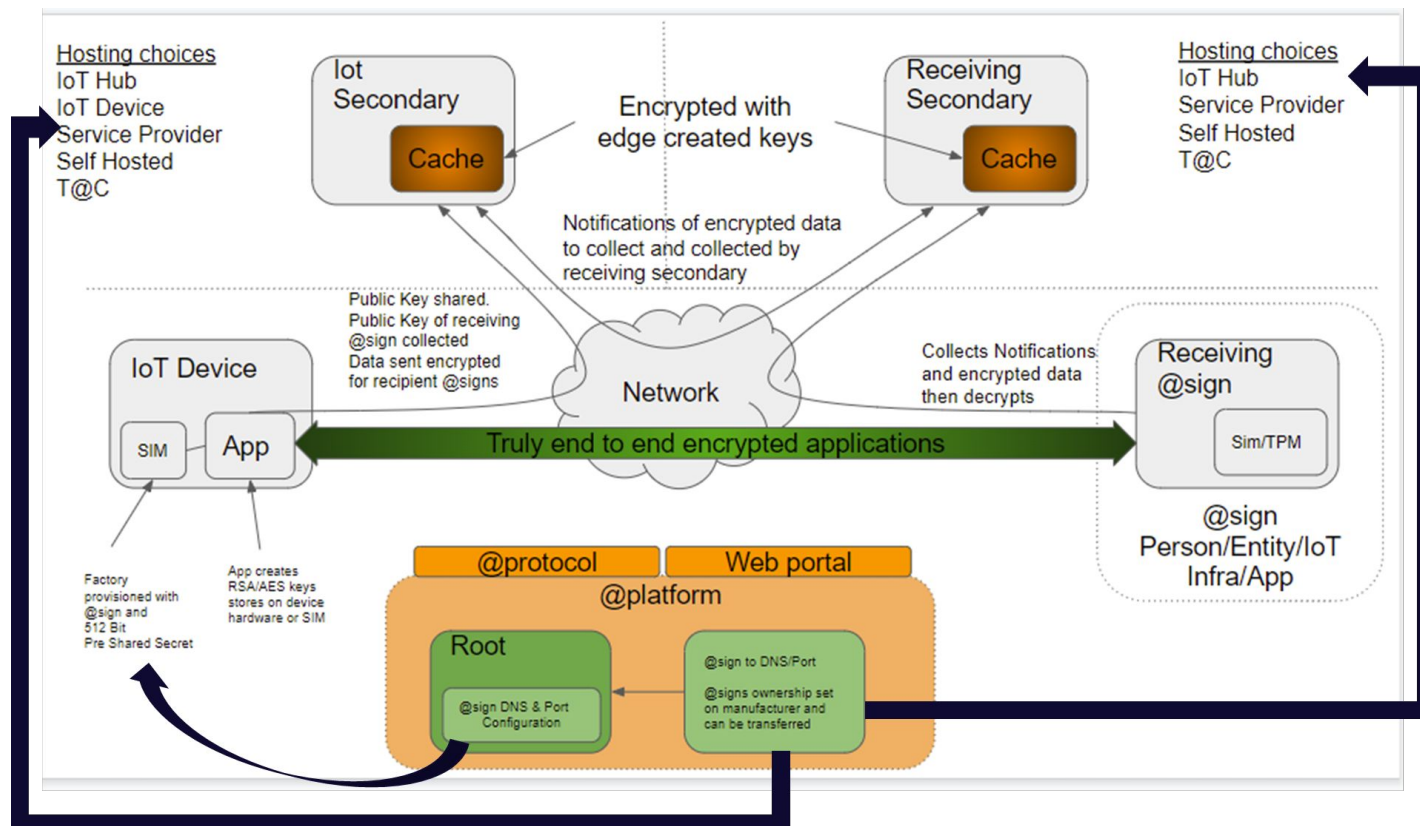


<https://morf.lv/implementing-pulse-oximeter-using-max30100>

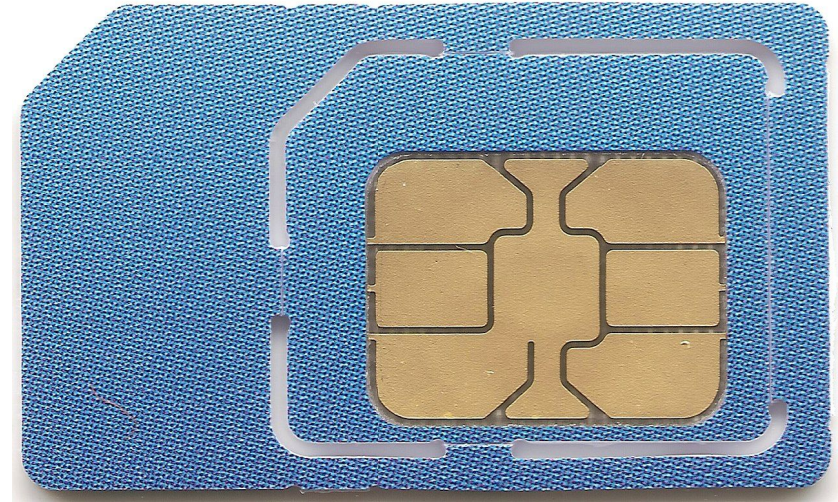


Connectivity, Security, Privacy

End to end encrypted



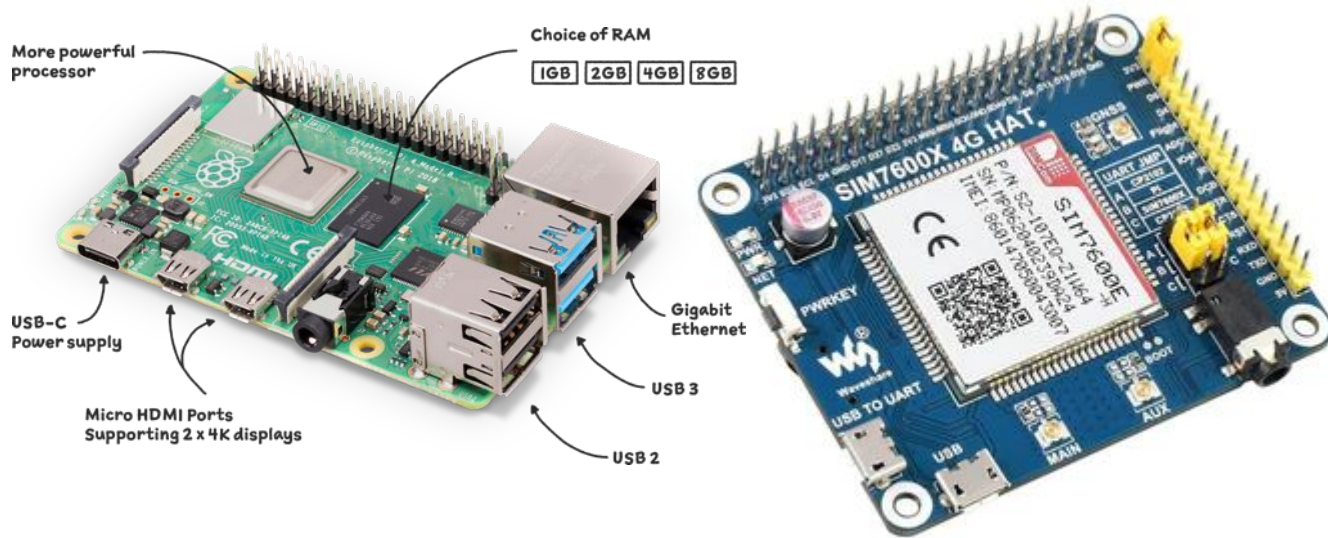
Where do the keys go?



[Keys by Jay Gooby](#) (CC BY 2.0)



Wired, WiFi and/or Cellular?



Admin access - Ssh! No ports



Ssh! No ports

ssh no ports provides a way to ssh to a remote linux host/device without that device having any open ports (not even 22) on external interfaces. All network connectivity is out bound and there is no need to know the IP address the device has been given. As long as the device has an IP address, DNS and Internet access, you will be able to connect to it.



<https://github.com/at-sign-foundation/sshnoports>

Local display
(the bit we didn't use Dart for)

PiTFT HAT presents as a framebuffer device



Review

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Takk skal du ha!

chris@atsign.com



@cpswan

Get started with the atPlatform



<https://atsign.dev/>



<https://github.com/atsign-foundation>



@atsigncompany

Join our  **DISCORD** community





Questions?